Marker cube mount: first-print check list

For Emily Simpson (3D print) and the team (bench test). Ben Zheng, 2026-11-09. Plain language on purpose. The goal is one printed part per probe (clip + stalk + cube), two copies each. Print one first, test it with this list, then print the other three.

Before printing

- [ ] Handle width and thickness measured on the real probe at the clip zone (100 to 130 mm above the probe face) and typed into the CAD file. The numbers in the file are estimates from a photo, about plus or minus 3 mm. On the L4-12t-RS the clip must sit above the two buttons.
- [ ] Stalk leans forward, over the patient, never towards the cable and the operator's wrist. Lean about 25 degrees. If more tilt clearance is needed make it more upright; a longer stalk does not help.
- [ ] Clamp style chosen. Two-piece clamp with M3 screws is the safe choice. A snap-fit C-clip is quicker but the opening has to be nearly as wide as the handle, so it holds less firmly and PLA may crack when flexed.
- [ ] Filament: white, matte. Not glossy, not grey, not "natural" translucent.
- [ ] Print settings: 0.2 mm layers or finer, 3 perimeters, 4 or more top layers, 10 to 15 percent infill. The cube is drawn solid so the slicer can hollow it.
- [ ] Orientation: no supports touching any of the five marker faces. The bottom face of the cube (where the stalk joins) and the clip can carry supports or sit on the bed.

When the part comes off the printer

1. **Flat faces.** Lay a steel ruler across each marker face. No daylight under it bigger than a sheet of paper (about 0.1 mm). Warped or bumpy faces make the tracker read the wrong angle.
2. **Recess.** A 51 mm sticker drops in and sits flush, not proud of the rim. If it is tight, the recess can grow by 0.5 mm. If it rattles, fine, the glue fixes it.
3. **Labels readable.** Face numbers 1 to 5 and the up-arrows can be read by eye. They tell us which marker went where.
4. **Fit on the probe.** With 1 mm foam tape inside, the clip closes on the handle and the probe can be shaken without the clip rotating or sliding. Buttons and cable are free. A normal scanning grip does not touch the clip or stalk, and the cable drapes back over the wrist without touching the stalk.
5. **Stiffness.** Push the cube sideways with a finger. It should flex less than a millimetre and spring back.
6. **Weight.** The part should weigh under about 80 g. The probe should not feel top-heavy in the hand.

On the bench (team, 20 minutes)

1. **Stickers on.** Print the marker sheet at 100 percent, measure a marker (35 mm), cut on the grey line, glue one per face. Write down which marker id went on which face number. Photograph the cube.
2. **Camera view.** Phone on the tripod about 30 cm away on the far side of the patient, looking down at 45 degrees towards the operator. In the normal grip the top face and the front face are both fully visible.
3. **Tilt clearance.** On the phantom, tilt the probe 45 degrees towards the cube side. The cube should stay about 7 cm above the surface (design value 74 mm; at 60 degrees about 24 mm remain). Nothing touches.
4. **Tracking.** Record a practice sweep. The tracker should find the cube in at least 95 percent of frames and the still probe should jitter by less than 1 mm. (2026-10-30 with the paper cube: 87 to 97 percent, 0.2 to 0.5 mm.)

5 **Numbers recorded.** Cube edge, marker edge, cube centre to probe face (from the CAD plus the measured clip height on the probe), marker-to-face map. These go into the layout file; nothing works without them.

If something fails

Problem	Likely cause	Fix
Clip loose or rotates	handle measured small, or corners rounder than modelled	reduce `foam` or `handle_w`/`handle_t` by 1 mm, thicker foam, reprint only the clip if it is a separate part
Clip will not close	handle measured large	add 1 mm to `handle_w`/`handle_t`
Face not flat	printed with supports on it, or top layers too few	reorient, more top layers, print cooler
Sticker proud of the rim	recess too shallow for the paper	`recess_depth` 0.7
Cube reaches the surface on a forward tilt	stalk leans too far forward	`stalk_lean` 20 (more upright); a longer stalk does not help
Cube hides behind the hand on camera	camera too low or too far to the side	raise the phone, move it to face the operator
Tracker loses the cube when the probe tilts	only one face visible	move the phone further behind the probe

After a pass

Print the remaining three (one more for this probe, two for the other probe with its own handle numbers). Keep the first print as the spare.